

ActInf Textbook Group ~ Cohort 2 ~ Meeting 20

(Chapter 9, part 1)

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okay welcome thanks it's uh March 15 2023 we're in meeting 20 of cohort two in our first discussion on chapter nine so as we begin this discussion of nine is there any General comment or question about nine that anyone wants to bring up one one part on nine is it at least points the direction towards an Avenue that many people who are interested in learning and applying active inference want to go down which is to make generative models yes that help us understand different cognitive systems and also have models that can accept empirical data so that we can enact the tale of two densities so we can use the model in the generative Direction to go from specified parameters to output data sets of synthetic data kind of like generative AI but then also to recognize outcomes in the world or data and then update or parameterize our generative models from those data ends it's that Tale of Two densities the generative density and the recognition density the ability to run these models both ways to generate valid synthetic data from a generative model and if you design your experiment so that those data which are generated are also actually being generated in the real world then you can just instantly plug it back in um and we'll I'm sure unpack and explore more so we'll go through the chapter but is there any just general comment or general thoughts on chapter nine first foreign so let's look at one previous question so how did chapter four on the generative model relate to chapter nine as we're getting close to the end of the book chapter six was a recipe it was about applying active inference chapter four we were learning the necessary prerequisites in chapter one and two and three were the roads to chapter four then chapter five we got the neurobiology and some examples chapter six we got the recipe chapter 7 and 8 was the discrete time and continuous time and now we're in chapter nine where we're learning how to solve problems and it's about getting one's hands dirty with case studies and problem solving we'll see this in the figure upcoming a key distinction is between the subjective model and the objective model there's different readings of the first sentence and in some readings it could be seen as playing fast and loose with instrumentalism and realism so the subjective

model though is that which the subject is that that which is subject specific from the subject's perspective now it turns out that it is a map that we're making from the subjects perspective so it isn't the actual territory of the subjectivity but it is from the subjects perspective and then to meet that and make the full model we are going to have an objective model which doesn't mean that it's the one and only it means that it's our view from the outside seeing that thing as an object this entire setup with a tale of two densities and the the metabasian enables us to make Bayesian cognitive models and think about experimentation in a Bayesian context which lets us talk about Bayes optimal cognition in different systems and one area that that has been especially relevant is in computational phenotyping this chapter deals with a utility of active inference formulations in analyzing data from behavioral experiments it's going beyond the proof of principle simulation so in the proof of principle domain we have the Jumping Frog in the hands we have the rat in the tmas and so on and here active inference is going to be used to answer scientific questions since the subjective GM is the generator of behavior in active inference then the scientific hypotheses that we're going to entertain about the causes and consequences of phenotype must be in terms of hypotheses about alternative generative models the challenge is to just like an abductive logic challenge generate a portfolio or spaces of alternative generative models cognitive models and then ask for a given string of observations what the evidence is for difference alternative hypotheses provided and then at that point one could take a more classical or frequentist approach they could also take a more pure Bayesian approach and ask about like the Bayes factor and the relative support of different models but the Upstream part is to generate alternative explicitly or implicitly by specifying State spaces or possibilities and then chapter 9 is when we're plugging in empirical data and then going to ask about which of those generative models or spaces within the possibilities of the GMS have what evidence with respect to the observables if you have any thought or questions just totally go for it otherwise we'll just continue to like walk through it on the first discussion Michael go for it yeah I'm uh I mean I guess you'll still come to it but I'm struggling with this mathematics uh very much and last time we had this interesting discussion about temporal models the difference between chapter 7 and chapter and chapter eight and I was thinking that for my data that I have with this gay State and the keystrokes and all this these temporal models would be perhaps much better Suited but then we discussed about these different uh kinds of derivations and I'm really Unsure how what that implies whether I should find a way to express my data in kinds of waves so that one can compute somehow derivations from this or um so we did discussed about The Disappearance of of time

in these models and I I'm really struggling to understand what that means in my data that I have but I don't know whether this really relates to what you were saying before yeah it was it's definitely it's it's um it's a it's a experience and a challenge when you have an actual uh data set and for people who are already experimentalists it's not like we can just dream up the most creative generative model and then build a lab to record those data in many cases were actually constrained in what we can do or we already have the data set and then we're going to work backwards from the data set to generative models yep that's the case in my case yes yeah yeah so I mean one um uh Beware all who pass this point is everything just like everything that led to the collection of your data was an experimentalist's choice you could have collected more or different um participants you could have done this or that in the data collection and now let's just say that we're in the analysis phase everything Downstream everything about the structure of the generative model and so on is all going to be modeler's degrees of freedom we are looking for different spaces to specify and then different portfolios of generative models to test against and one of the great things about Bayesian statistics is that wildly disparate models architecturally can be compared through the base factor and the Bayesian information Criterion so just a explain why that's relevant relative to the um frequency statistics When comparing frequentist statistical models one of the only principled ways to actually test for which model is better is the hierarchical likelihood ratio test so you can say I have two strictly nested models I have factor a and Factor B and now I'm going to test for an interaction effect of a x b and then you can ask whether there's a statistically significant Improvement by taking on that interaction effect but if you said I have a model with a XB and then I have a model with B and C and C and D and DNA and X and Y those models cannot be directly compared because they're likelihoods are not comparable in contrast Bayesian models can be compared in various ways the base Factor the Bic accuracy minus complexity free energy even when there's a very very different model anatomies so it's not even like you have to decide whether you want to do only a discrete or only a continuous time generative and hopefully as the software pipelines become more developed it'll be like you'll specify all the different channels that your data can flow through and then see how diff wildly different match up with the data and look at their relative evidence or the areas that they fit well or don't in and you still might have to prune that down in some talk or paper but also maybe it'll be possible to share those computational resources also with that kind of high bandwidth um but yeah we'll we'll return to it so two reasons why we wanna fit data like why chapter nine the first is to estimate parameters of interest from a group those kinds of computational phenotypes might be useful

for some other setting the second reason is to compare alternative hypotheses so here we have in the simplest case two groups and we want to ask if there is a difference in some parameter in the cognitive model a related reason to use computational phenotyping is potentially even for one person or one group or it could be for two groups is to compare two different models against one another and ask which one has different um evidence and so these are the two agendas estimating parameters given a cognitive and evaluation among models given data that's the same hypothesis you have the same hypothesis you just replace the model same data so like uh like uh let's just say in the first case what this would look like would be people who have condition X or not I'm going to measure their attention with with this given attention model I'm going to test if there's a significant difference in this attention parameter this one would say I have two attention models I have one where attention is a single layer model and I have one where there's like a you know a nested model on attention and now on the whole data set or on just this one person I'm going to ask which of these two cognitive models is more consistent with the data that I see and and this um these two agendas are perfectly compatible with different terms Within Bayes theorem which is a tight synthesis of the kind of Foley Bayesian epistemology it's like we're doing Bayesian science with Bayesian statistics we're approaching our data collection and Analysis in an optimal information setting okay now we get to figure nine one with the objective and the subjective so does anyone want to give a thought on figure 9.1 what they see here or what they see this figure doing in the context of the book okay just to give a first thought then again just raise your hand or write in the chat or anything that we saw in chapters four five seven eight all those generative models all those base graphs those are in the center in the dashed line and so it's a map but it's a map that we're constructing as if it were a view from the inside so here's the mouse in the T Maze and here's the pomdp in this case it's it's a discrete time you can tell because it has the past present and future transition Matrix and so on and so we've made a discrete time model of the mouse in the teammates so it's the as if Mouse's View From the Inside and and that's what makes sense when we're doing computational phenotyping we don't want to do model selection on um omniscient mice because we don't think that that exists in the world so the kinds of constraints that we think actually are uh justifiable in the world like the only input is sensory and the only outputs are actions those constraints are embodied within the GM that we make for the subject but the subject is also the object of our study the object of our study or the system of interest of our study is that Mouse in the teammates and so with respect to how we designed that teamaze experiment and whether we do it with one trial on one day or we do it with this other

block design or we have it replicated across Laboratories those experimental parameters that are action selections from us can be understood as being drawn from distributions of possible what are action selections by the experimenter are being passed into the subjects model as an observation what is passed out of the subject through action is the input for the Observer so we're kind of wrapping the generative models that we saw in chapter four Etc calling those subjective models not to mean that they're willy-nilly but just to say that they are as if from the subject's perspective and then wrapping that as a behavioral researcher in an objective model not to mean that it's the one only true model but it's the one from the outside with the subject as object and this is called meta Bayesian so in some ways this is a formalization of ethology with the cognitive modeling twist because we're explicitly saying that we're interested in this cognitive or internal map Michael has has this to do with um self-evidencing experimental that they put in the data and they get the results and then they mother the subject in between in such a way that the expectations become true that the input output relations and so is it has it to do with itself evidencing ideas yeah the experimental yeah it's very interesting um like our subjective model we can see why that is set up in a way to be self-evidencing we can say that the mouse is moving around in the team A's so that it reduces Divergence so that it can be the kind of mouse that it expects and prefers to be we can understand its diverse behavioral Selections in light of self-evidencing if we're going to consider ourselves in a Bayesian light as well then also that entails a certain kind of epistemological self-evidencing now in the extreme or in one simple reading self-evidencing would be like you write down the number four oh I wonder what number four great write down the number four look at the number it's four well that isn't what self-evidencing means anymore than for the organism it means to stay in the dark room which has been a point of philosophical contention for like literally 10 years people say well if organisms are acting to reduce surprise or to self-evidence why don't they just stay in the dark room dark room challenge you know Throw Down The Gauntlet and it's like right if the generative model didn't have survival requirements and its only imperative was to have the reduced Divergence with a photon distribution you would and you do see that behavior but that's not what organisms do and Maps aren't territories so if you actually built a realistic generative you would find it self-evidencing doesn't just mean staying in the dark room it also means self-evidencing for temperature and thirst and all these other drives so self-evidencing by the researcher plays out in a way that would need its own generative model to be specific about but for example if the researcher says my commitment is to maximum information experimentation I'm not trying to Simply confirm what my friend told me my commitments

and expectations preference is for maximally informative experiment design then the experiment that they design can be understood as self-evidencing even if the data points are maximally surprising that would be the point Ali and also about self-evidencing is in recent literature especially since 2019 there's two distinct way of formulating uh self-evident self-evidencing one is in terms of Bayesian model evidence which is the approach taken uh in this chapter as well and one is in the the terms of lagrangian of the autonomous Paths of active particles so these autonomous paths are actually or uh in other words these lagrangians are actually the expected green energy which can by themselves decomposed into two separate terms corresponding with expected costs on one hand and expected Information Gain so uh in this kind of formulation the expected cost is the surprisal of sensory paths which is also what's referred to as uh the sensory surprisal or the self self evidencing of the self-organization systems so uh there are slight uh or there's some subtle differences between these two formulations but ultimately they converge onto same results providing I mean a description of the behavior of these systems in uh or in particular active particles uh in the environment in terms of minimizing their expected thank you when we set up the ethological moment in this way it's an active inference problem so it can be approached using all of the amazing tools that are being developed our focus is on active and specifically discrete time however the generic methods may be used with other likelihood functions so other normative models of behavior in active inference our normativity is around self-evidencing things should persistent things should minimize or bound their surprise but there's no issue with having a subjective model with a different normativity that it should maximize reward as per a utility function defined now there will be a generic inference scheme that can be used for metabasian inference then there's going to be a recipe and then a key example readers uninterested in certain details are invited to skip 9.39.4 okay we'll keep that in mind parametric empirical base is a technique that's used to jump start a Bayesian analysis with empirical data so a common question by adherence and critics in Bayesian statistics is like well what priors should I choose because there's no such thing as a unbiased or a non-biased prior there's the uniform distribution like the flat probability distribution but even that is over a certain interval and it still embodies a prior albeit one that is uniform across the range but there's other situations where the uniform distribution is not even a possible selection for example you can't have a uniform distribution across all possible Heights of trees at some point you just got to be like you know objects have this range of sizes and so there's always this question about selection of priors and Hyper priors and and so on now in many cases in practice it can be shown that it doesn't matter that

much for example somebody will show that with two wildly Divergent prior distributions there's so much signal in the data that it converges to the same posterior so that's often used as evidence like hey we tried different priors and we always got the same conversions even with different families of priors so we feel like this signal is being driven by patterns in the data and it's not an aberration because of our choice of like a very opinionated prior so that's one strategy people take presentation of alternative priors another approach that can be taken and it's not exclusive to what was previously mentioned is parametric empirical base and so in this situation empirical data are used to kick start setting the priors so we're going to be measuring the height of children in the whole school first we're going to measure one classroom and then we're going to use the mean and the variance of that classroom as our prior moving forward for the other classrooms so it's just a way to take empirical data and use the empirical data summary like mean invariance for gaussian to jump start subsequent Bayesian analyzes including of the data that generated that prior now that analysis by itself won't be super informative because you kind of already described the maximum likelihood solution for that one data set but with sub sampling and so on there's some ways to make PEB really useful all right so 9.3 9.4 it was like reminded that they're not essential so we'll move a little faster through them but we can return to it next time here we have the variables that are being used in the metabasian image in 9 figure 9.1 and there's a focus on the likelihood function and um Bayesian model evidence closely linked to surprise or self-evidising because the model with the most evidence is the one that is least surprised by so sometimes we're talking more about maximizing likelihood other times minimizing the surprise but the tuning of the parameters is not going to clash between those approaches π_i are those policy selections and so π_i is going to be based upon a minimization of free energy over all the π_i s so this is like active free energy minimization of the generative model not of the territory of the GM is going to be used as the imperative for Action selection which is what we observed from The Mouse and the teammates so if we have the model that does best at predicting their behavior then we've done a good compromised job on the subjective perception cognition action and there's just a mention of this temperature parameter shaky handedness which is just kind of fun to like think about even just from an intuition perspective where in the shaky hand limit the high temperature everything is shaking a lot it eradicates differences in the likelihood of taking different actions like if something is just vibrating it just ends up making all action selection choices equally likely whereas like in the sort of um short-handed limit you follow either your habits or your free energy minimization without further hindrance sure handed optimization of

policy shaky handed if and if it's soft Max is one then you're sampling from the posterior so there's two actions and your posterior evaluation is 60 40. if this parameter is one you you select those policies sixty and forty percent of the time respectively if you turn up the temperature it converges back to 50 50 and if you turn down the temperature you're always going to select the higher um likelihood one there are some details on the LaPlace um assumption there's probably a lot of detail that could be gone into but LaPlace puts a quadratic function that is censored at the maximum likelihood solution to a distribution so if it's bimodal there's two humps to the distribution LaPlace will get stuck at one of them or you'll get other kinds of issues like it'll be stuck at one but very over dispersed and so on so this isn't approximating the distribution completely but it turns out it's a super computable way that in many reasonable situations does really well now we return to the PEB we want to ask regular Anova type statistical modeling questions here we have a linear model this is kind of like SPM Style where X is a design Matrix it's also used in linear regressor models won't go into this in too much more detail but this is kind of like saying just like you could do a linear regression with a design Matrix on height you can do a linear regression with a design Matrix on generative model parameters and still test for an effect of age by height on attention or something like that 9.5 instructions for model based analysis Ollie go for it yeah I just wanted to point out um the relation between the uh LaPlace assumption and the form of the recognition density because um very briefly uh assuming I mean LaPlace assumption uh mathematically leads to this assumption that that the shape of the recognition density would be necessarily a gaussian density so in that case the variation of free energy would be equivalent to Gibbs energy but without that LaPlace assumption then we would have a variational free energy equals Gibbs energy minus the entropy of the recognition density and by recognition density I mean the probability that the approximate probability density for hidden inputs um hidden input variables under a certain generative model many times getting to a gaussian or a quadratic simplifying it's symmetric it has the minimum number of parameters too to describe like the mean and the scaling the mean the variance but that doesn't mean distributions in the world are like that however for ones that are even roughly shaped like that like they have a central tendency these kinds of symmetric models can be like super super effective because they have very very well established methods for solving now we're going to get to instructions for the model based analysis so definitely contrast slash juxtapose this with the recipe from chapter six so in six we were kind of coming to the system what are we modeling why are we doing it what structurally are we looking to include in our GM how are we going

to specifically set up that GM and then how are we going to set up the generative process okay doesn't have to be in order but these are what we'll need to figure out now in chapter 9 we have some instructions for how to go from not just building a GM like we saw in six but to using the GM for GM based analysis collecting behavioral data including metadata or demographic data two formulate a pomdp I think here they could have said construct a GM as per chapter 6. but they're focusing like they mentioned above on the discrete time case so that's why they're focusing on the pomdp specify a likelihood function that is going to simulate behavior and quantify the likelihood of observed actions we started with data then we formed a pomdp basically like it was laid out in chapter six it turns out we're doing a pmdp like chapter seven but it's basically like formulating the GM and now we're going to actually be using for doing the kinds of papers that people write about where this age group had with this statistical confidence level a higher likelihood of this or how should the model be used to calculate a likelihood specification of priors that's where you can take a zero centered approach or again one could use to just say oh previous studies found this distribution so that's going to be my prior distribution here's our prior probability and our likelihood what we need now is the posterior probability in the model evidence likelihood function prior solving for now with data posterior distributions and evidence we put information in on the front half of Bayes equation now we're solving for the second half of Base equation and there are routines in SPM in Matlab that help with this lastly perform the group level analysis this is also an opportunity for using PEB or you can do any other kind of truly wide variety of statistical methods you could fall back to frequentism and just do a t-test just say all right I had 17 in this group and 35 in this group and we got a t-test to see if this attention parameter was significantly different you could use a canonical variate analysis or an independent contrast analysis or principal component analysis any statistical method that you would see in any paper this is just doing statistics at this point a little bit more detail and figure 9.2 is showing the six step inversion procedure collect the data make the model specify the likelihood function P of u given θ remember back to 9.1 until does the observed Behavior so we're doing behavioral modeling what we're interested in is the distribution of probabilities of behaviors given θ and m so now we've kind of like specified the pomdp for the mouse in the maze but we've pulled back our likelihood to in a way abstract or insulate from all of those subjective model parameters such that our statistical analysis is only about objective model parameters so I'd be like if we were making a cognitive model of a decision to take X or Y Behavior we'd want to do that then collect the data on the behavior and then do a demographic

analysis on how X or Y is related with this or that and those statistics are not an appeal to the cognitive model of the decision maker they're actually a layer of descriptive that will be absolutely familiar to anyone who's done behavioral modeling in the lab it's just a slightly different way of looking at them but it can be seen as doing model evaluation and estimation using statistics in a way that doesn't mention any of the inner cognitive model parameters because things have been framed in this nice tan metabasian way we can do the tale of two densities and here is kind of a visualization of what the um t-test would look like those those error bars could be bootstrapped et cetera et cetera but let's just imagine that they're from a t-test so here's five and here we have some parameter that we modeled their novelty seeking something or some other specific model inside and then we can ask like is there more variance within or among These Bars and maybe you see you know a bar with an asterisk here these two are indistinguishable but they're both significantly different than these three and you know this one is significantly different from that one but not that so you see those kinds of like regular Anova type statistics here in a Bayesian setup so specification of the GM chapter 6 and all that allows us to do statistics on it just if you had some Advanced model of materials properties and then we were just testing which one was like harder or softer before analyzing actual data we may want to check face validity by generating fictive data so taking just a look at it visually and also doing parameter recovery so that means can you identify or recover parameters or do you have issues with your model like collinearity where there's two parameters that are like they're so covariate with each other that sometimes you'll run the model and it will explain the data equally well with wildly different combinations of those two parameters just for example um and again classical SPM like angle with a design Matrix it's kind of amazing how it works but the design Matrix encodes all of the experimental metadata and then you have like a matrix of regressors and you just smash the design Matrix into this beta and somehow the dimensions work out and it's like what you'd want to know like in SPM it'll be one Matrix with this is the um the age and so on of the of the patients and here's a big Matrix with all of the fmri data and you just smash them together and it and then it sets up this if it's all set up properly okay and then just we'll look at these upcoming sections but then talk more with open questions and and look at any other details there's going to be an analysis of the Mirza at all socket eye movement smooth pursuit of eye movement atoms and mersa smooth tracking of eye movement on the left atoms discrete tracking of eye movements mersa not much more said than that obviously there's papers to go back to and those are even generative models that we can use replicate them and modify them models of false inference

so this is going to be approaching that kind of challenging Nexus of well if it's Bayes optimal then how is their maladaptive Behavior how can people have delusions How can there be the accommodation of chemical changes that that just fundamentally alter the substrate of cognition table 9.1 has a sample of examples that um our papers exploring computational pathology in primarily humans but almost surely vertebrates addiction impulsivity compulsivity delusions hallucination interpersonal and personality disorder ocular motor syndrome pharmacotherapy prefrontal syndromes visual neglect disorders of interceptive inference so just giving us some examples and the notes tend to be defining what that pathology was operationalized as by those studies and then briefly what kind of generative models were used to support what new explanations predictions or control opportunities are presented when we have for example an active model of oculomotor syndrome as opposed to some other framework for describing that behavior a useful way of thinking about causes of pathological behaviors to think about the prior belief used for policies and how each part of this may be disrupted to give rise to abnormal policy selection so one could look at a figure and just imagine that we're dealing with a very simple uh mouse-like patient on the top here or um just looking at it here if we're interested in that behavioral policy selection component that is uh conditioned on preferences and affordances or habits so that's why they mentioned c and e because maybe there's a group of people who are repeatedly or compulsively taking a given Behavior but again just totally simplifying making this up there might be some who prefer that behavior and understand the consequences of their action so they engage in the selection of the behavior because they expect slash prefer it in another case it actually might not be aligned with the preferences but it has become habitual in which case there's a difference cognitive model parameterization and potentially different consequences or treatment opportunities however the experiment needs to be carefully designed to resolve those possibilities which is why a great advance is to be able to basically not just pre-register what you're going to do but to do the statistical power analysis and ask which regions of experimental State space are going to be Bayesian optimal for for example distinguishing two groups of people some of whom have a habitual bias and others have a strong prior preference for something but it's not just like any given set of experimental data are going to resolve this because these are not measurable things in the world they're inferred parameters in our cognitive map so it's not just going to happen because we tried that being said we can make the GM ahead of time and do the simulations do that parameter identification and Recovery to determine what kinds of experimental paradigms could resolve that if that was what we were trying to resolve um then

there's a few more uh interesting points here on everything from salience and neuromodulators and different pathologies and they summarize they outlined an approach to use the previously described to pose questions to empirical data so chapter 6 was like the recipe for making the GM and then here we saw like the kind of second stage of the recipe the instructions for model based that helped us see like not just the building of the model which got us to but now we see the whole six stages that gets us like to the visualization and statistical analysis that we want in the paper um they also highlighted that you can use these methods to distinguish groups for example with a given cognitive model or you can use it to distinguish alternative model hypotheses given data from one or more group they gave a bunch of examples in table nine one and they focused on the examples of atoms and Mirza at all in eye tracking Pursuit giving just a total clear example of how even for the very same you can model it with continuous or it's a generic method for experimental design that offers an opportunity to answer questions about the function of the nervous system and health and disease that's good for our first discussion on nine we'll come back next week and check any questions that people have added and uh yeah thanks we'll come back next week